

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0908

Garnet Mesa 138 kV
August 03, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0908. The Interconnection Customer proposes a 123.7 MW solar photovoltaic facility interconnecting at the Garnet Mesa 138 kV substation in the Pennsylvania area. The request was received on December 24, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 321 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$24,714,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Sugar Hill – Allegheny Energy Units 3 4 & 5 345kV, the Yucca – Fraley 138kV Line loads to 142.3% of its 383 MVA rating. Project DFAX: 10.6%.

For the loss of Allegheny Energy Units 3 4 & 5 – Sugar Hill 345kV, the Fraley Substation – Houseville 138kV loads to 141.4% of its 391 MVA rating. Project DFAX: 10.6%.

The Va Hospital – Houseville 138kV L is overloaded to 141.1% of its 395 MVA emergency rating for the loss of the Sugarhill – Allegheny Energy Units 3 4 & 5 345kV. The project contributes 10.6% DFAX (13.1 MW) to this overload.

The Sugar Hill – Va Hospital 138kV line is overloaded to 140.4% of its 403 MVA emergency rating for the loss of the Sugar Hill – Allegheny Energy Units 3 4 & 5 345kV. The project contributes 10.6% DFAX (13.1 MW) to this overload.

For the loss of Sugar Hill – Ritter Park 345kV, the Garnet Mesa – Mt. Vernon 138kV loads to 139.3% of its 187 MVA rating. Project DFAX: 44.0%.

The Yucca – Maricao 6301 Substation 138kV is overloaded to 134.5% of its 189 MVA emergency rating for the loss of the Yucca – Regal 138kV line. The project contributes 6.0% DFAX (7.4 MW) to this overload.

The Maricao 6301 – Station J 138kV is overloaded to 134.2% of its 184 MVA emergency rating for the loss of the Regal – Yucca 138kV. The project contributes 6.0% DFAX (7.4 MW) to this overload.

For the loss of Sugar Hill – Ritter Park 345kV, the Mt. Vernon – Buffalo Rd 138kV loads to 124.0% of its 203 MVA rating. Project DFAX: 44.0%.

For the loss of Yucca – Regal 138kV L, the Station J – Webbers Falls 138kV loads to 121.3% of its 195 MVA rating. Project DFAX: 6.0%.

Regal – Sloss Industries Corp 138kV: 119.0% loading (220 MVA) following the outage of the Allegheny Energy Units 3 4 & 5 – Southeast Vicksburg 500kV; DFAX = 5.4%.

Southeast Vicksburg 345 kV/138 kV Autotransformer: 118.9% loading (376 MVA) following the outage of the Southeast Vicksburg – Allegheny Energy Units 3 4 & 5 500kV; DFAX = 5.2%.

Chadwicks – Buffalo Rd 138kV Line: 118.2% loading (209 MVA) following the outage of the Ritter Park – Sugar Hill 345kV L; DFAX = 44.0%.

Sloss Industries Corp – Southeast Vicksburg Substation 138kV Line: 115.4% loading (224 MVA) following the outage of the Southeast Vicksburg – Allegheny Energy Units 3 4 & 5 500kV; DFAX = 5.4%.

The Webbers Falls – Lake Goodwin 138kV is overloaded to 112.4% of its 205 MVA emergency rating for the loss of the Yucca – Regal Substation 138kV. The project contributes 6.0% DFAX (7.4 MW) to this overload.

Sugar Hill 345/138 kV TX: 111.8% loading (465 MVA) following the outage of the Sugar Hill – Ritter Park 345kV; DFAX = 19.4%.

The Yucca Substation – Regal 138kV is overloaded to 109.7% of its 246 MVA emergency rating for the loss of the Allegheny Energy Units 3 4 & 5 Substation – Southeast Vicksburg 500kV. The project contributes 5.4% DFAX (6.7 MW) to this overload.

Columbus-Rapelje Auto – Chadwicks 138kV Line: 105.9% loading (222 MVA) following the outage of the Ritter Park – Sugar Hill 345kV; DFAX = 44.0%.

For the loss of Allegheny Energy Units 3 4 & 5 – Sugarhill 345kV Line, the Siletz – United Coen Inc 138kV loads to 101.8% of its 174 MVA rating. Project DFAX: 8.8%.

For the loss of Yucca Substation – Regal 138kV line, the Lake Goodwin – Allegheny No. 5 Hydro Station 138kV Line loads to 101.4% of its 217 MVA rating. Project DFAX: 6.0%.

Little Chute – Fort Humbug 345kV: 100.0% loading (1,720 MVA) following the outage of the Coronet – Ft Humbug 500kV; DFAX = 5.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yucca – Fraley 138kV Line	Rebuild line with higher-capacity conductor	\$3,310,000	16% (\$543,000)
Fraley – Houseville 138kV L	Rebuild line with higher-capacity conductor	\$2,696,000	16% (\$442,000)
Va Hospital – Houseville 138kV	Rebuild line with higher-capacity conductor	\$2,530,000	16% (\$415,000)
Sugar Hill – Va Hospital 138kV line	Rebuild line with higher-capacity conductor	\$4,956,000	16% (\$813,000)
Mt. Vernon Substation – Garnet Mesa 138kV	Rebuild line with higher-capacity conductor	\$3,273,000	62% (\$2,012,000)
Yucca – Maricao 6301 138kV Line	Rebuild line with higher-capacity conductor	\$5,198,000	16% (\$836,000)
Maricao 6301 – Sta J 138kV line	Rebuild line with higher-capacity conductor	\$1,433,000	16% (\$230,000)
Mt. Vernon – Buffalo Rd 138kV	Rebuild line with higher-capacity conductor	\$3,665,000	62% (\$2,253,000)
Webbers Falls – Sta J 138kV line	Rebuild line with higher-capacity conductor	\$3,671,000	16% (\$590,000)
Regal – Sloss Industries Corp 138kV	Rebuild line with higher-capacity conductor	\$4,822,000	15% (\$741,000)
Southeast Vicksburg 345 kV/138 kV Autotransformer	Replace transformer with higher-rated unit	\$15,034,000	20% (\$2,961,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chadwicks – Buffalo Road 138kV	Rebuild line with higher-capacity conductor	\$4,592,000	62% (\$2,823,000)
Sloss Industries Corp – Southeast Vicksburg Substation 138kV Line	Rebuild line with higher-capacity conductor	\$4,761,000	15% (\$731,000)
Lake Goodwin – Webbers Falls 138kV line	Rebuild line with higher-capacity conductor	\$4,148,000	16% (\$667,000)
Sugar Hill 345/138 kV TX	Replace transformer with higher-rated unit	\$24,783,000	16% (\$3,943,000)
Yucca Substation – Regal 138kV	Reconductor line	\$876,000	15% (\$135,000)
Columbus-Rapelje Auto – Chadwicks 138kV	Reconductor line	\$3,463,000	62% (\$2,129,000)
Siletz – United Coen Inc 138kV	Reconductor line	\$348,000	16% (\$55,000)
Allegheny No. 5 Hydro Station – Lake Goodwin 138kV	Reconductor line	\$4,210,000	16% (\$677,000)
Little Chute – Fort Humbug 345kV	Reconductor line	\$59,448,000	3% (\$1,718,000)
	Total allocated to project		\$24,714,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.